



Broadband High Power Amplifier Module (SSPA)

REV.3.0_2024.11.18

6-18GHz/10Watt/Module

Model Number: KB60180M40B

The model KB60180M40B is a multi-octave high power amplifier operating between 6 GHz and 18 GHz and offering a wide dynamic range with 10 Watts typical saturated power. The employment of gallium nitride (GaN) and chip-and-wire technology in manufacturing ensures this module state-of-the-art power performance with excellent power-to-volume ratio. It is ideal for jamming, EMC, test and measurement applications.

FEATURES:

- Small size and light weight
- Instantaneous ultra-broadband
- 50 Ohms input and output matched
- Built-in control and protection circuits
- Class AB

ELECTRICAL SPECIFICATIONS @ +28.0VDC, 25°C, 50Ω

Parameter	Symbol	Minimum	Typical	Maximum	Units
Operating Frequency	BW	6000		18000	MHz
RF Output Power @P _{SAT}	P _{SAT}		10		Watt
Power Gain @P _{SAT}	G _p		40		dB
Power Gain Flatness	ΔG _p		±2		dB
Input Return Loss	S ₁₁			-10	dB
Harmonics @5W	H		-20		dBc
Spurious Signals	Spur		-60	-55	dBc
In/Output Impedance	Impedance		50		Ω
Operating Voltage	V _{DC}	26	28	30	Volt
Power add efficiency	Eff		10		%
DC Current @10W	I _{DD}		3		Amp
Switching Time @1kHz TTL	T _{on/off}		2	5	uSec

MECHANICAL SPECIFICATIONS

Parameter	Value	Units	Limits
Dimension	150x90x25 [5.91x3.54x0.98]	mm [inch]	Maximum
Weight	0.8 [1.76]	kg [lbs]	Maximum
RF Connector Input	SMA, Female		
RF Connector Output	SMA, Female		
DC Interface Connector	D-Sub 9-Pin, Male		
Cooling	External Heatsink (Not Supplied)		

ENVIRONMENTAL CHARACTERISTICS (Design to Meet)

Parameter	Minimum	Typical	Maximum	Units	Notes
Operating Temperature	-40		60	°C	
Non-operating Temperature	-45		85	°C	Storage
Relative Humidity (non-condensing)			95	%	

Absolute Maximum Rating

Input RF drive level without damage	+10 dBm	Maximum
Load VSWR @ P _{OUT} =5W	∞ @ all load phase & amplitude for duration of 1 minute; 3:1 @ all load phase & amplitude continuous	
Over Temperature	85°C @ heatsink [restored @ 60°C]	

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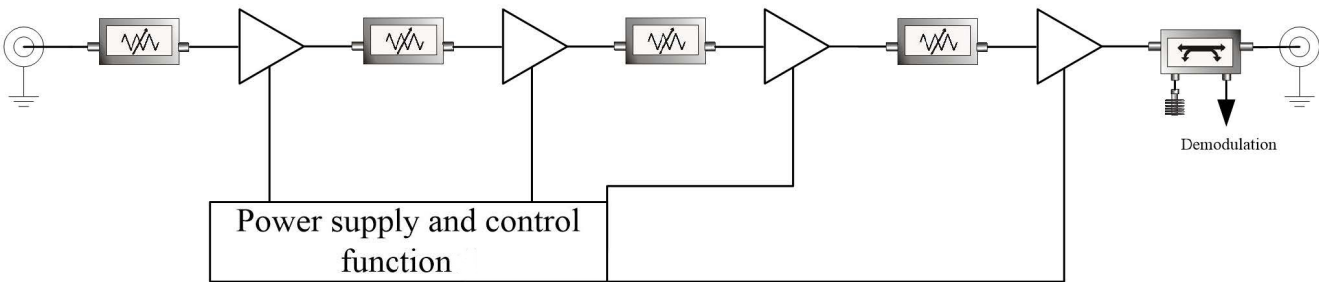
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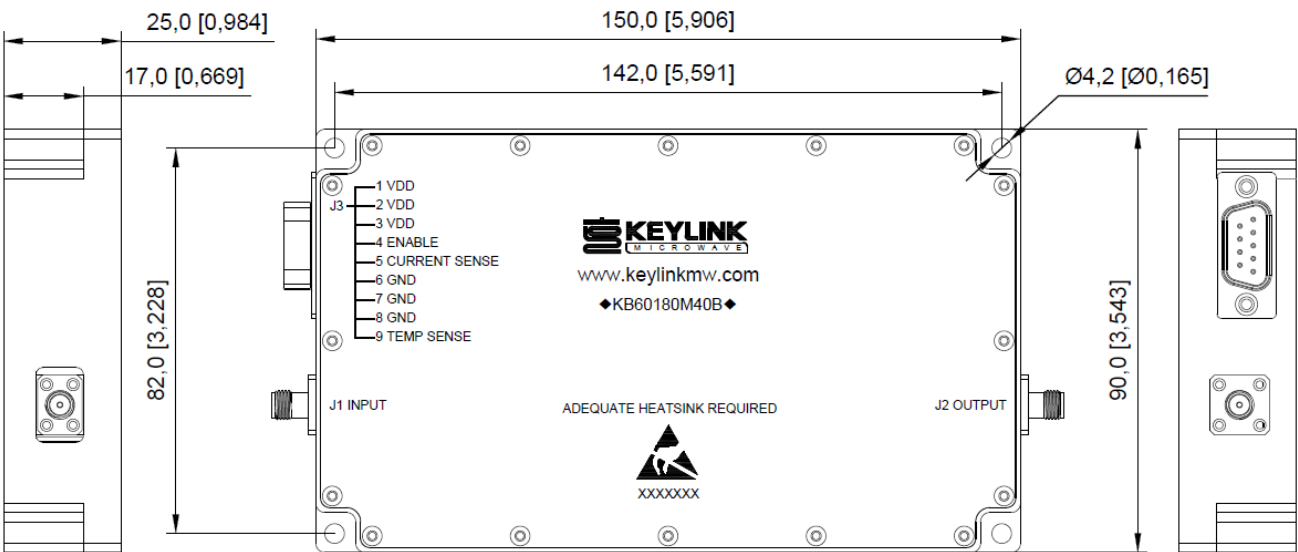
DC INTERFACE CONNECTOR

Pin #	Description	Specifications
1	VDD	28VDC
2	VDD	28VDC
3	VDD	28VDC
4	ENABLE	Amplifier Enable: TTL Logic High (3.3V) (Internally Pulled-Low)
5	CURRENT SENSE	Analog voltage relative to IDD @ 100mV per Ampere
6	GND	Ground
7	GND	Ground
8	GND	Ground
9	TEMP SENSE	Analog voltage relative to Module's Temperature @ 10 mV/°C

Functional Diagram



OUTLINE DRAWING [All dimensions in mm (inch)]





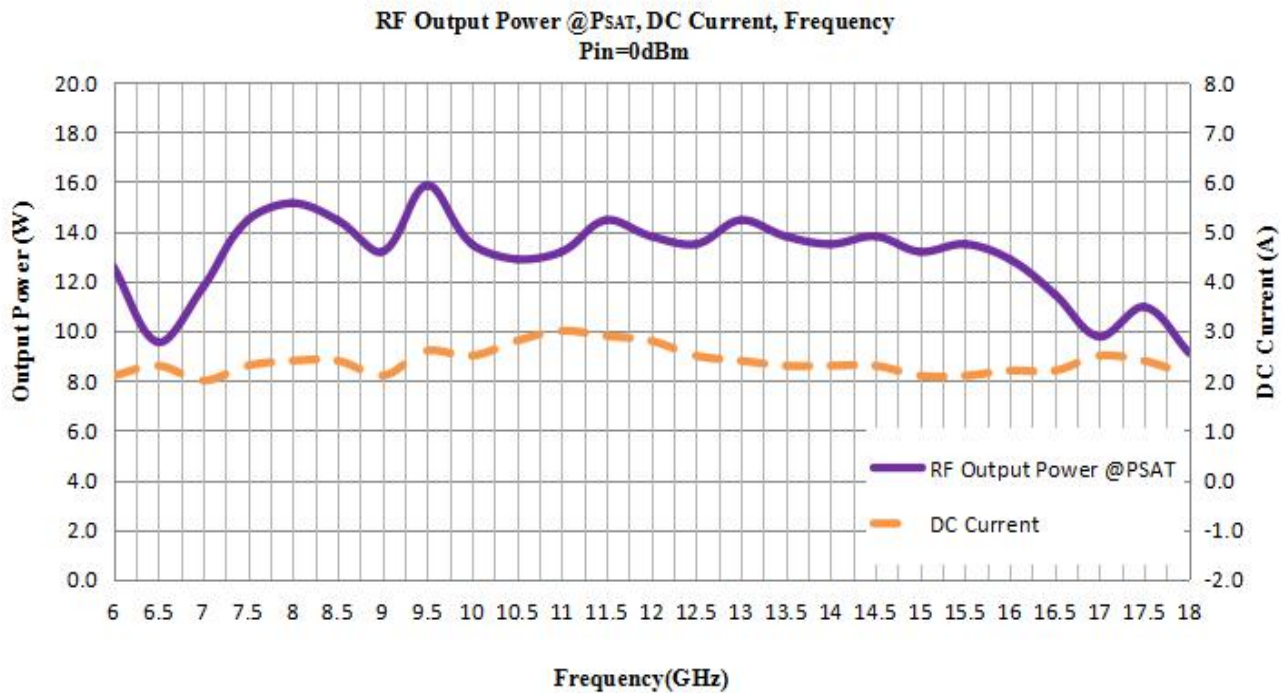
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TYPICAL PERFORMANCE PLOTS (For reference only)



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